2.8 증명

① 증명 $a^x \equiv 1 \pmod{m}$ then $\underbrace{d}_{\mathrm{ord}(a)} \mid x$

∵ · otherwise $x = dq + r$ with $d \in \mathbb{Z}$

$0 \le r < d$

· $1 \equiv a^x \equiv (a^d)^q \cdot (a^r)$

$\equiv 1 \cdot (a^r) \pmod{m}$

→← d is the least natural number with $a^d \equiv 1 \pmod{m}$

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② 증명 $\underbrace{\mathrm{ord}(a^k)}_{= d'} = \dfrac{d}{(d, k)}$ with $d = \mathrm{ord}(a) \pmod{m}$

∵ suppose $(a^k)^x \equiv 1 \pmod{m}$

firstly $\dfrac{d}{(d, k)} \le x$ because $d \mid kx$ (①번에 의해)

$\dfrac{d}{(d, k)} \mid x$

moreover $(a^k)^{\frac{d}{(d,k)}} \equiv (a^d)^{\frac{k}{(d,k)}} \equiv 1 \pmod{m}$

inside $\mathbb{Z}$

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③ $\overset{k}{\mathrm{ord}(ab)} = \{\overset{r}{\mathrm{ord}(a)}, \overset{s}{\mathrm{ord}(b)}\}$

if $(\mathrm{ord}(a), \mathrm{ord}(b)) = 1 \pmod m$

$k \mid rs$ $\quad (ab)^{rs} \equiv (a^r)^s (b^s)^r$

$\uparrow$ $\quad \equiv 1^s \cdot 1^r \equiv 1 \pmod m$

$k = rs$

$\downarrow$

$rs \mid k$ $\quad (ab)^{kr} \equiv a^{kr} b^{kr}$

$\equiv (1)^k b^{kr} \equiv 1 \pmod m$

$\Rightarrow$ $s \mid kr$ therefore $s \mid k$ (s와 r은 서로소)

$\Rightarrow$ likewise $r \mid k$

$\Rightarrow$ $rs \mid k$ (s와 r은 서로소)

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④ • odd prime p

• primitive root $g \pmod{p^j}$ ← 무조건 존재함

Claim either g or $g + p^j$ becomes primitive root $(\mathrm{mod}\ 2 \cdot p^j)$

∵ • 홀수 선택 G then $(G, 2p^j)$

• let G belongs to $d \pmod{2p^j}$

• $G^d \equiv 1 \pmod{2p^j}$ hence $d \mid \phi(2p^j)$

• $G^d \equiv 1 \pmod{p^j}$ hence $\phi(p^j) \mid d$

• $\phi(p^j) = \phi(2p^j)$ therefore $d = \phi(2p^j)$.

2. Find a primitive root of 23.

$p = 23, \quad \phi(p) = 22$

$5^1 \equiv 5 \quad (23) \quad \Rightarrow \quad 5^{22} \equiv 1 \quad (23)$

$5^2 \equiv 2 \quad (23)$

therefore 5 is primitive root.

$5^{11} \equiv 2^5 \cdot 5$

$\equiv 160$

$\equiv -1 \quad (23)$

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8. Use Theorem 2.37 to determine how many solutions each of the following congruences has:

(*a*) $x^{12} \equiv 16 \pmod{17}$ (*b*) $x^{48} \equiv 9 \pmod{17}$

(*c*) $x^{20} \equiv 13 \pmod{17}$ (*d*) $x^{11} \equiv 9 \pmod{17}$.

3 is primitive root of 17.

$3^1 \equiv 3$

$3^2 \equiv 9$

$3^3 \equiv 10$

$3^4 \equiv 13$

$3^5 \equiv 5$

$3^6 \equiv 15$

$3^7 \equiv 11$

$3^8 \equiv 16 \quad (\text{mod } 17)$

(a) $12 \text{ ind } x \equiv \text{ind } 16 \quad (\text{mod } 16)$

$12 \text{ ind} x \equiv 8 \quad (\text{mod } 16)$

$3 \text{ ind} x \equiv 2 \quad (\text{mod } 4)$

$\text{ind } x \equiv 2 \quad (\text{mod } 4)$

$x \equiv 3^2, 3^6, 3^{10}, 3^{14} \quad (\text{mod } 17)$

$\equiv 9, 15, 8, 2 \quad (\text{mod } 17)$

(b) $48 \text{ ind } x \equiv \text{ind } 9 \quad (\text{mod } 16)$

$48 \text{ ind} x \equiv 2 \quad (\text{mod } 16)$

$\Rightarrow$ no solution as $(48, 16) \nmid 2$

$= 16$

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(*c*) $x^{20} \equiv 13 \pmod{17}$ (*d*) $x^{11} \equiv 9 \pmod{17}$.

(c)

$$20 \operatorname{ind}(x) \equiv \operatorname{ind}(13)$$

$$\equiv 4 \pmod{16}$$

$$5 \operatorname{ind}(x) \equiv 1 \pmod{4}$$

$$\operatorname{ind}(x) \equiv 1 \pmod{4}$$

$$x \equiv 3^1, 3^5, 3^9, 3^{13}$$

$$\equiv 3, 5, 14, 12 \pmod{17}$$

(d)

$$11 \operatorname{ind} x \equiv \operatorname{ind} 9 \pmod{16}$$

$$\equiv 2 \pmod{16}$$

$$\operatorname{ind} x \equiv 6 \pmod{16}$$

$$x \equiv 3^6 \equiv 15 \pmod{17}$$

4. Find the values of $\left(\frac{a}{p}\right)$ in each of the 12 cases, $a = -1, 2, -2, 3$ and $p = 11, 13, 17$.

- $\left(\frac{-1}{p}\right) = \begin{cases} 1 & p \equiv 1 \pmod 4 \\ -1 & p \equiv 3 \pmod 4 \end{cases}$

- $\left(\frac{2}{p}\right) \equiv \begin{cases} 1 & p \equiv 1 \text{ or } 7 \pmod 8 \\ -1 & p \equiv 3 \text{ or } 5 \pmod 8 \end{cases}$

- $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$

- $\left(\frac{3}{p}\right) = (-1)^{\frac{p-1}{2}} \left(\frac{p}{3}\right)$

$p \backslash a$	-1	2	-2
11	-1	-1	1
13	1	-1	-1
17	1	1	1

- $\left(\frac{3}{11}\right) = (-1)^{\frac{11-1}{2}} \left(\frac{11}{3}\right) = (-1)^5 \left(\frac{2}{3}\right)$

  $= (-1) \times (-1) = 1$

- $\left(\frac{3}{13}\right) = (-1)^{\frac{13-1}{2}} \left(\frac{13}{3}\right) = (-1)^6 \left(\frac{1}{3}\right)$

  $= (1)(1) = 1$

- $\left(\frac{3}{17}\right) = (-1)^{\frac{17-1}{2}} \left(\frac{17}{3}\right) = (-1)^8 \left(\frac{2}{3}\right) = (1) \times (-1) = -1$

13. Prove that if r is a quadratic residue modulo $m > 2$, then $r^{\phi(m)/2} \equiv 1 \pmod{m}$. (H)

if $x^2 \equiv r \pmod{m > 2}$

then $r^{\phi(m)/2} \equiv 1 \pmod{m}$

$\because$ ① $(r, m) = 1$ therefore $(x, m) = 1$

② $x^{\phi(m)} \equiv 1 \pmod{m}$ by Euler

③ $\phi(m)$ is even because $m > 2$

④ $1 \equiv (x^2)^{\phi(m)/2}$

$\equiv r^{\phi(m)/2}$

***19.** For all primes p prove that $x^8 \equiv 16 \pmod{p}$ is solvable. (H)

Note. $0^8 \equiv 16 \pmod{2}$

$p > 2$ then either $\left(\frac{2}{p}\right) = 1$ or $\left(\frac{-2}{p}\right) = 1$ or $\left(\frac{-1}{p}\right) = 1$

case ① $c^2 \equiv 2 \pmod{p}$

case ② $c^2 \equiv -2 \pmod{p}$

case ③ $c^2 \equiv -1 \pmod{p}$

$\Rightarrow$ $c^4 \equiv 4 \pmod{p}$

$c^8 \equiv 16 \pmod{p}$

$\ulcorner$ $x = 1 + c \ (p)$ then

$$x^2 \equiv (1+c)^2$$
$$\equiv 1 + 2c + (-1)$$
$$\equiv 2c \pmod{p}$$

$$x^4 \equiv 4c^2$$
$$\equiv -4 \pmod{p}$$

$$x^8 \equiv 16 \pmod{p}$$

***20.** Let p be an odd prime. Prove that if there is an integer x such that

$p \mid (x^2+1)$ then $p \equiv 1 \pmod 4$;

$p \mid (x^2-2)$ then $p \equiv 1$ or $7 \pmod 8$;

$p \mid (x^2+2)$ then $p \equiv 1$ or $3 \pmod 8$;

$p \mid (x^4+1)$ then $p \equiv 1 \pmod 8$.

Show that there are infinitely many primes of each of the forms $8n+1, 8n+3, 8n+5, 8n+7$. (H)

Euler $\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod p$ therefore $\frac{p-1}{2}$ even

$= 1$ $\qquad p \equiv 1 \pmod 4$

$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$ therefore $p \equiv \pm 1 \pmod 8$

$= 1$

$\left(\frac{-2}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{2}{p}\right)$

$= 1$

case ① $\left(\frac{-1}{p}\right) = 1 = \left(\frac{2}{p}\right)$ $\Rightarrow$ $p \equiv 1 \pmod 8$

$p \equiv 1 \pmod 4$ $\qquad p \equiv 1, 7 \pmod 8$

case ② $\left(\frac{-1}{p}\right) = -1 = \left(\frac{2}{p}\right)$ $\Rightarrow$ $p \equiv 3 \pmod 8$

$p \equiv 3 \pmod 4$ $\qquad p \equiv 3, 5 \pmod 8$

$x^4 \equiv -1 \pmod p$ therefore order$(x) \in \{1, 2, 4, 8\}$

$x^8 \equiv 1 \pmod p$ $\qquad x^4 \equiv 1 \ (p)$ $\qquad$ order should be 8.

$\rightarrow\leftarrow$

$x^{p-1} \equiv 1 \pmod p$

$8 \mid p-1$ (Fermat)

***20.** Let p be an odd prime. Prove that if there is an integer x such that

$$p \mid (x^2+1) \text{ then } p \equiv 1 \pmod 4;$$

$$p \mid (x^2-2) \text{ then } p \equiv 1 \text{ or } 7 \pmod 8;$$

$$p \mid (x^2+2) \text{ then } p \equiv 1 \text{ or } 3 \pmod 8;$$

$$p \mid (x^4+1) \text{ then } p \equiv 1 \pmod 8.$$

Show that there are infinitely many primes of each of the forms $8n+1, 8n+3, 8n+5, 8n+7$. (H)

각 형태의 소수가 k개 존재한다면 .... with $P = p_1 \cdots p_k$

$8n+1$ $\quad N = (2P)^4 + 1 \Rightarrow \underbrace{q}_{\text{소인수}} \mid (2P)^4 + 1$ with $q \equiv 1 \pmod 8$ (단 q)

$\Rightarrow \forall j \quad (2P)^4 + 1 \equiv 1 \pmod{p_j}$

therefore $q \neq p_j$

$8n+7$ $\quad N = 8P^2 - 1 \Rightarrow$ 홀수들의 곱 $P^2 \equiv 1 \pmod 8$

$N \equiv 7 \pmod 8$

- 만약 N의 모든 소인수 $q \equiv 1 \pmod 8$ 이면 $N \equiv 1 \pmod 8$

  $\Rightarrow N$은 $q \not\equiv 1 \pmod 8$ 인 소인수를 가진다.

- $q \mid N$이면 $q \mid 2N = (4P)^2 - 2$

  이므로 (성질 ②) $q \equiv 1$ or $7 \pmod 8$

- $q \equiv 7 \pmod 8$ 인 소인수가 있다.

- $q \neq p_j \quad (\forall j)$

• $8n+3$ $N = p^2+2$ $\Rightarrow$ 홀수들의 곱 $p^2 \equiv 1 \pmod 8$

$N \equiv 3 \pmod 8$

• 만약 N의 모든 소인수 $q \equiv 1 \pmod 8$ 이면 $N \equiv 1 \pmod 8$

$\Rightarrow$ N은 $q \not\equiv 1 \pmod 8$ 인 소인수를 가진다.

• $q \mid p^2+2$ 이면 (성질 ②) $q \equiv 1$ or $3 \pmod 8$

$\Rightarrow$ $q \equiv 3$ 인 소인수가 있다.

. $q \neq p_j$ $(\forall j)$

$8n+5$ $N = (2p)^2+1$ $\Rightarrow$ 홀수들의 곱 $p^2 \equiv 1 \pmod 8$

$= 4p^2+1$ $N \equiv 5 \pmod 8$

• 만약 N의 모든 소인수 $q \equiv 1 \pmod 8$ 이면 $N \equiv 1 \pmod 8$

$\Rightarrow$ N은 $q \not\equiv 1 \pmod 8$ 인 소인수를 가진다.

• $q \mid (2p)^2+1$ 이므로 (성질 ①) $q \equiv 1, 5 \pmod 8$

$\Rightarrow$ $q \equiv 5 \pmod 8$ 인 소인수가 있다.

. $q \neq p_j$ $(\forall j)$

***22.** Show that if p and q are primes, $p = 2q + 1$, and $0 < m < (p + 1)^{1/2}$, then m is a primitive root (mod p) if and only if it is a quadratic nonresidue (mod p).

- prime p, q
- $p = 2q + 1$ $(p \geq 5)$
- $0 < m < \sqrt{p+1}$

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m is primitive root $(\text{mod } p)$ $\Longleftrightarrow$ m is quadratic non-residue

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$(\Rightarrow)$

- m belongs to $2q$
- since $m^{2q} \equiv 1 \pmod p$

$$m^q \equiv \underbrace{m^{\frac{p-1}{2}} \equiv -1}_{} \pmod p$$

by Euler m is quadratic non-residue.

$(\Leftarrow)$

- by Euler $m^{(p-1)/2} \equiv m^q \equiv -1 \pmod p$
- order(m) can be $1, 2, q, 2q$

① if $m^1 \equiv 1 \ (p)$ then $m^q \equiv 1 \ (p)$ $\rightarrow\leftarrow$

② if $m^q \equiv 1 \ (p)$ $\rightarrow\leftarrow$

③ if $m^2 \equiv 1 \ (p)$ then $m \equiv \pm 1 \pmod p$

$m \equiv -1 \pmod p$

$0 < m < p$ then $m = p-1$

$0 < m < \sqrt{p+1}$ therefore $p - 1 < \sqrt{p+1}$

$p^2 - 2p + 1 < p + 1$

$p^2 - 3p < 0$

$0 < p < 3$, $p = 2$. $\rightarrow\leftarrow$ $p \geq 5$

$\Rightarrow$ order$(m) = 2q$